Hierarchical Moc
RNA-seq

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Hierarchical Models for RNA-seq

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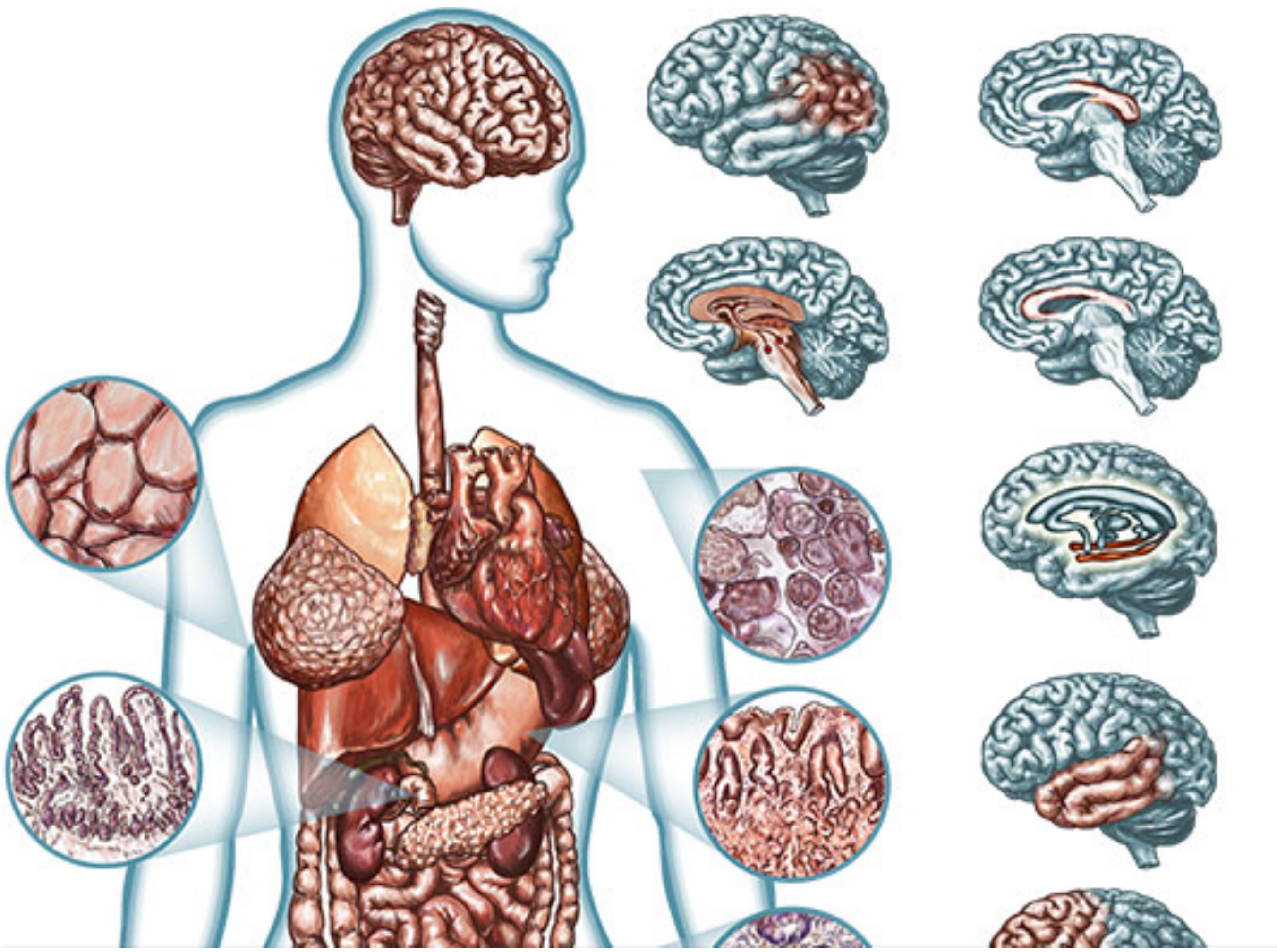
Outline

- Why RNA
- Motivation of hier. model for continuous
- Extension to sequence count data
- Questions

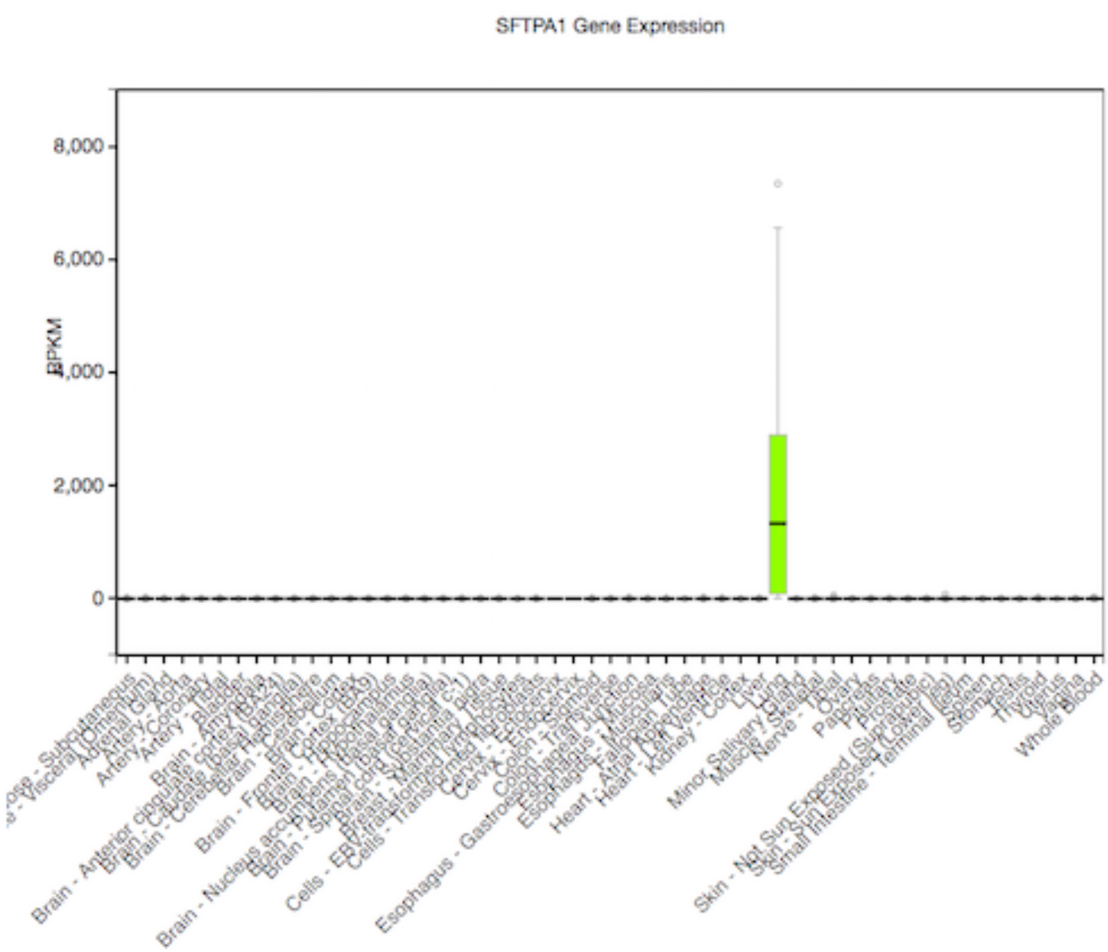
Why measure RNA as phenotype

Why measure RNA: tissue diversi



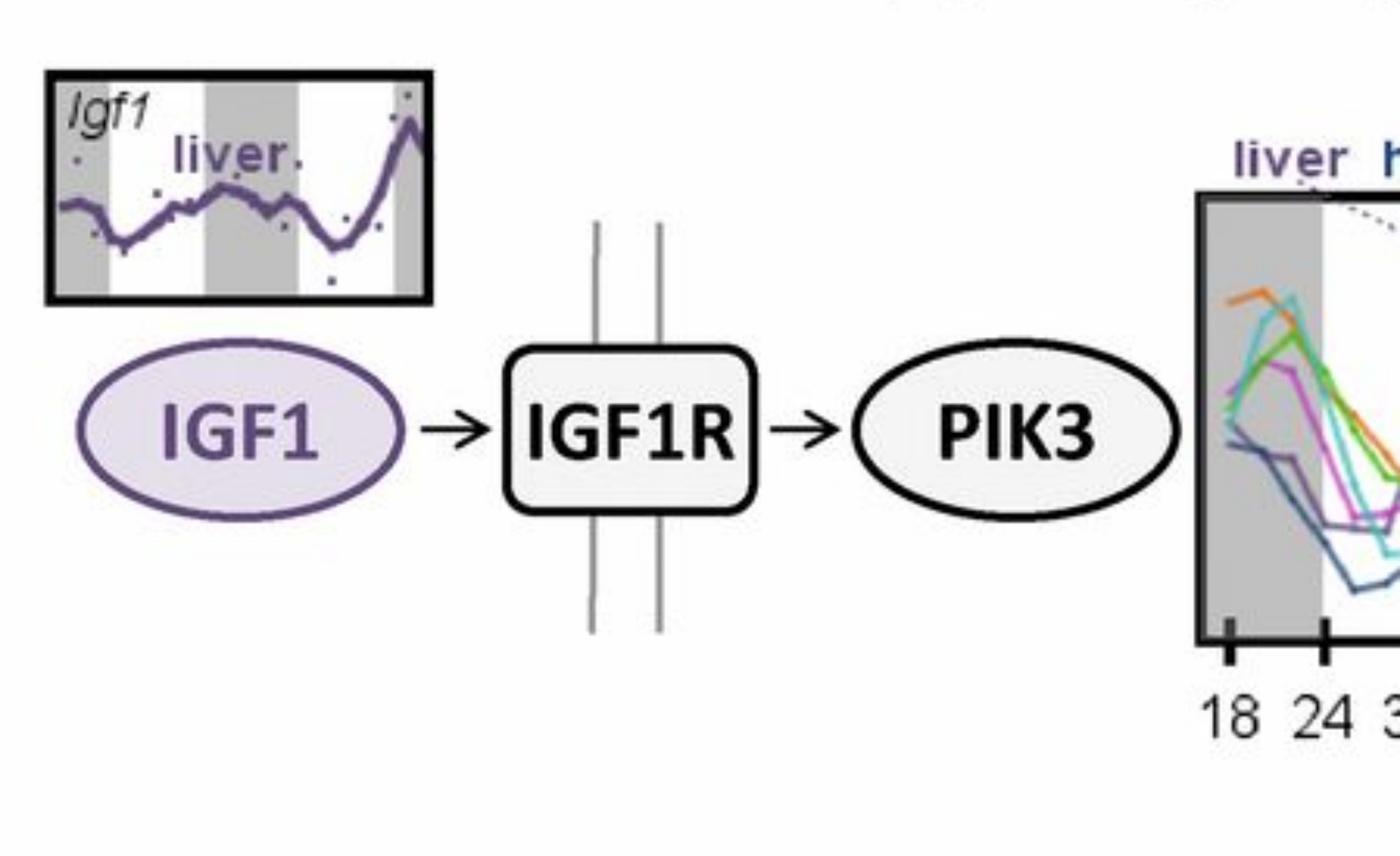
Roadmap Epigenomics

Why measure RNA: tissue diversi



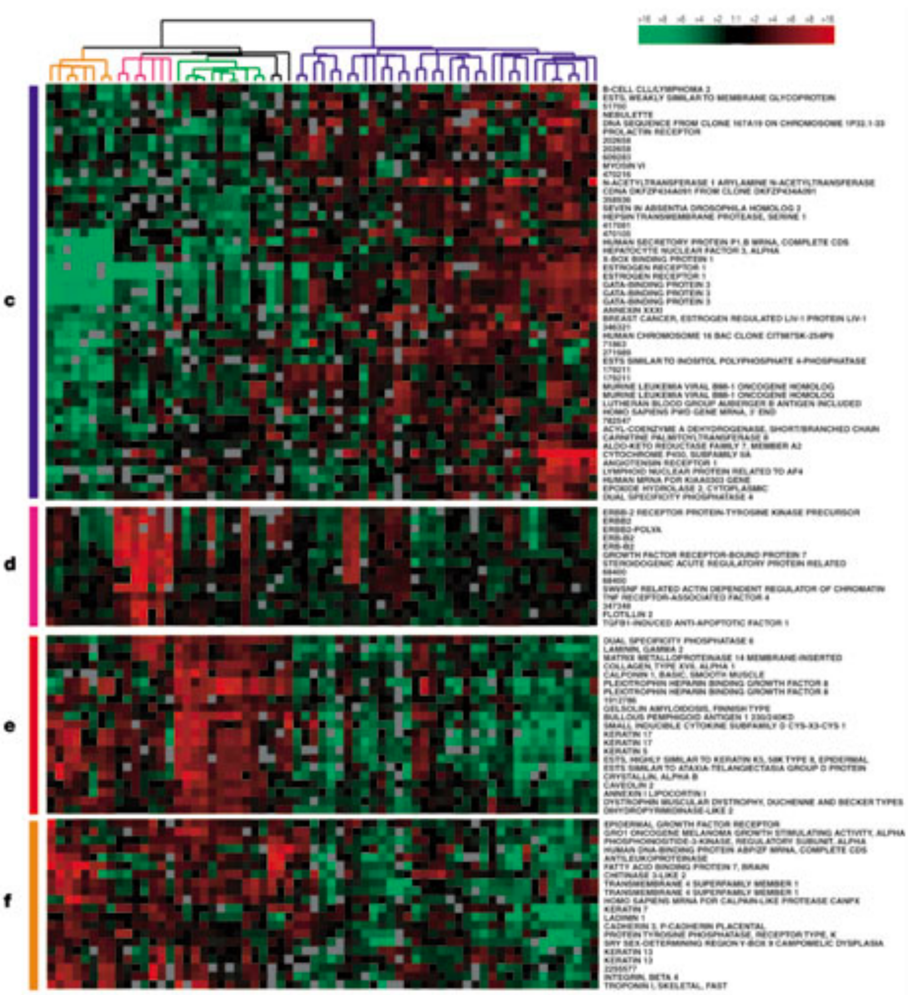
GTEx

Why measure RNA: within tissue



Zhang, et al. Circadian gene expression atlas (2014)

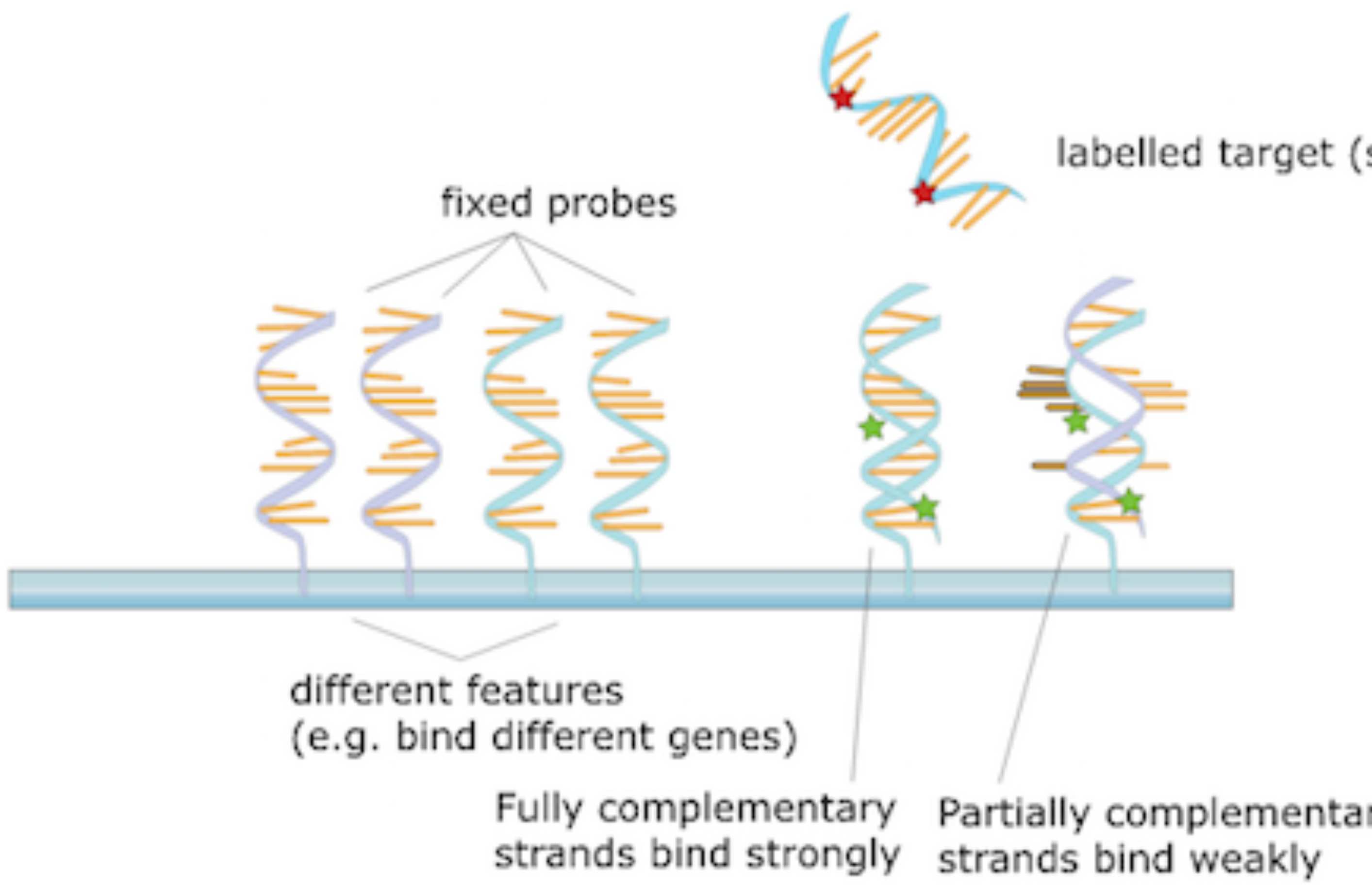
Why measure RNA: disease sub-t



Perou, et al. Molecular portraits of human breast tumours (2000)

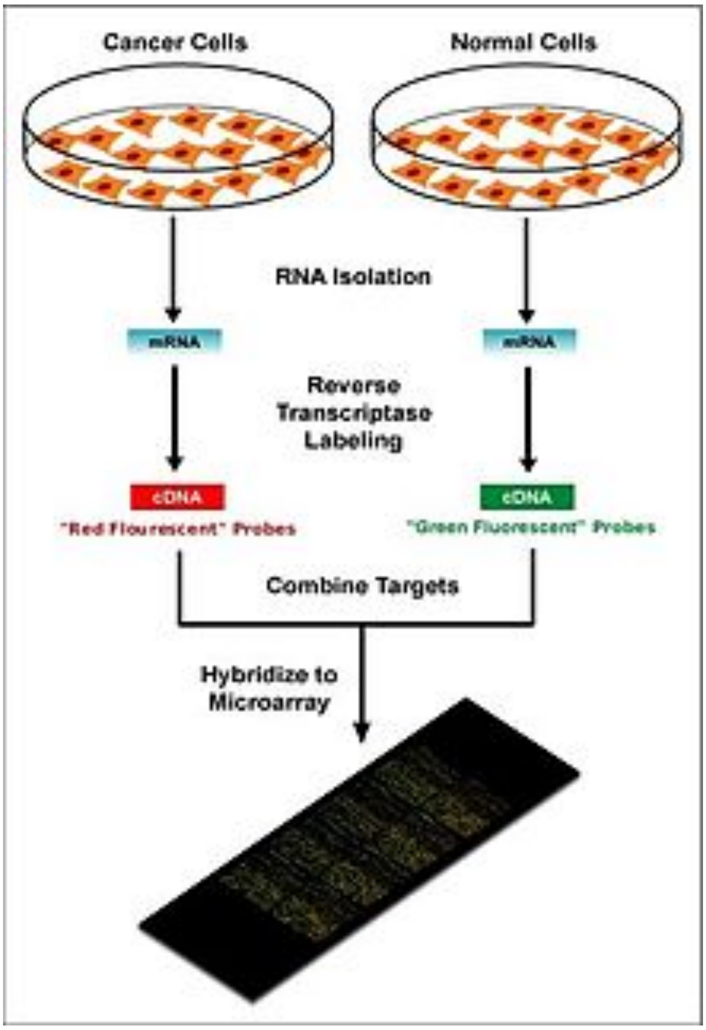
Step back: pre-sequencing

Before sequencing was microarray



Step back: pre-sequencing

Signal was captured light (positive, “continu



Formalize

- Formalize as a simple statistical model
- Observed data X_{ij} as a R.V. following a di
- Distributional parameters of interest

Motivating problem

- Gene expression for i=1,…,N genes and j=
- log of "true" expression values (non-nega
- Suppose 2 equal sized groups of samples

$$X_{ij} \sim N(\mu_{ij}, \sigma_i)$$
$$\mu_{ij} = \mu_{i0}, \quad j \in A$$
$$\mu_{ij} = \mu_{i0} + \delta_i, \quad j \in B$$

$\delta_i \neq 0$ implies DE (differential expression)

Note σ_i

Different genes *i* have different amount of v

$$
\begin{aligned}
X_{ij} &\sim N(\mu_{ij}, \sigma_i) \\
\mu_{ij} &= \mu_{i0}, \quad j \in A \\
\mu_{ij} &= \mu_{i0} + \delta_i, \quad j \in
\end{aligned}
$$

Goal of differential expression

- Find a set of genes for which $\delta_i \neq 0$
- We want to target a false discovery rate (F
- For genes in our set G at FDR threshold z

$$E\left(\sum_{i \in G} 1_{\{\delta_i = 0\}}\right) \leq |G$$

Is this realistic?

- Can we accomplish this if all $\delta_i \neq 0$
  - no, because methods often rely on glol
    normalization
- Are any $\delta_i = 0$?
  - maybe not, but many are very small fo
    experiment

Is this realistic?

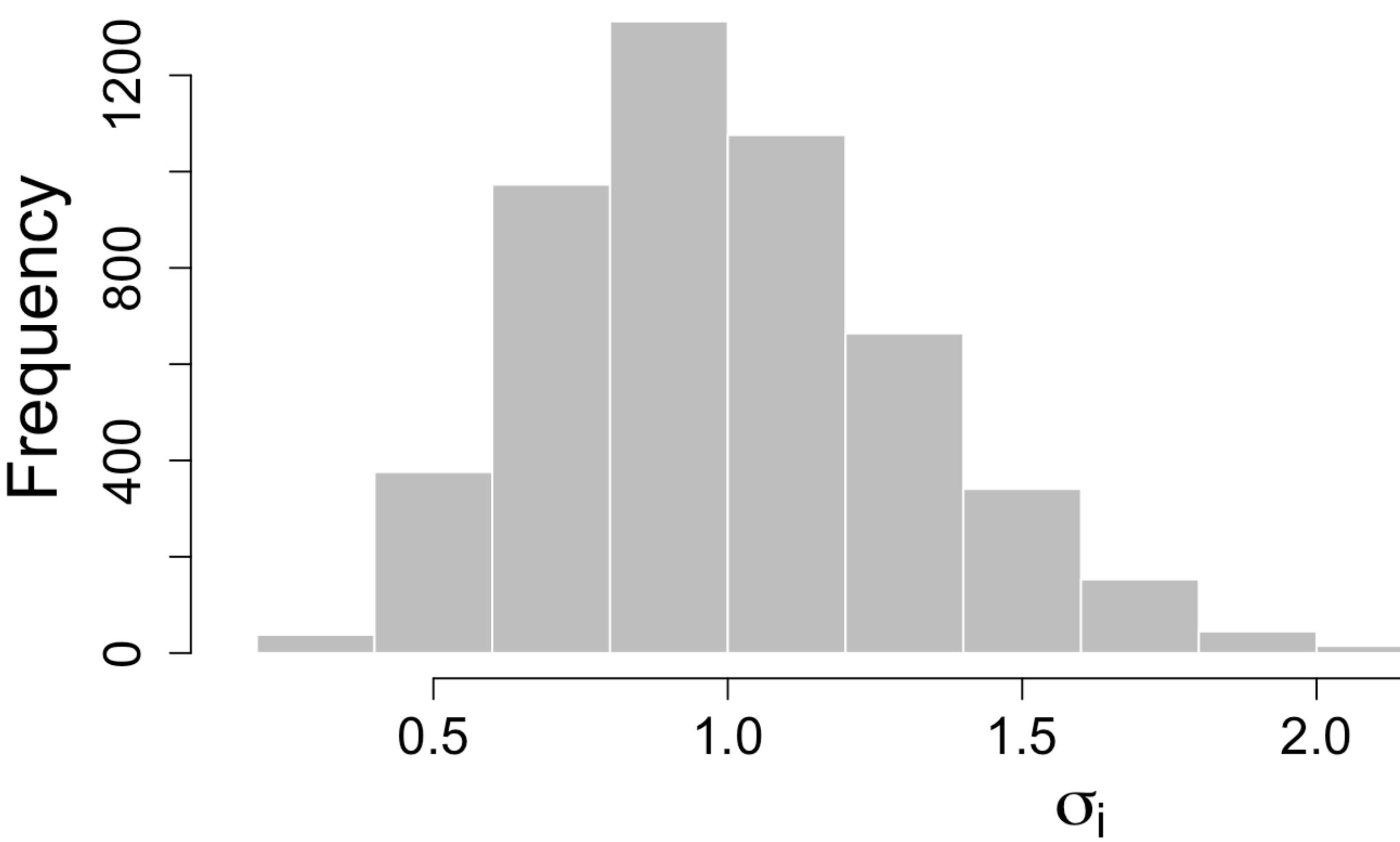
- What about σ_i for both groups?
  - often this is enough, larger variance do
  - not for single cell experiments
- More complex parametric models: baySe
- Non-parametric: SAM / SAMseq

Back to the model

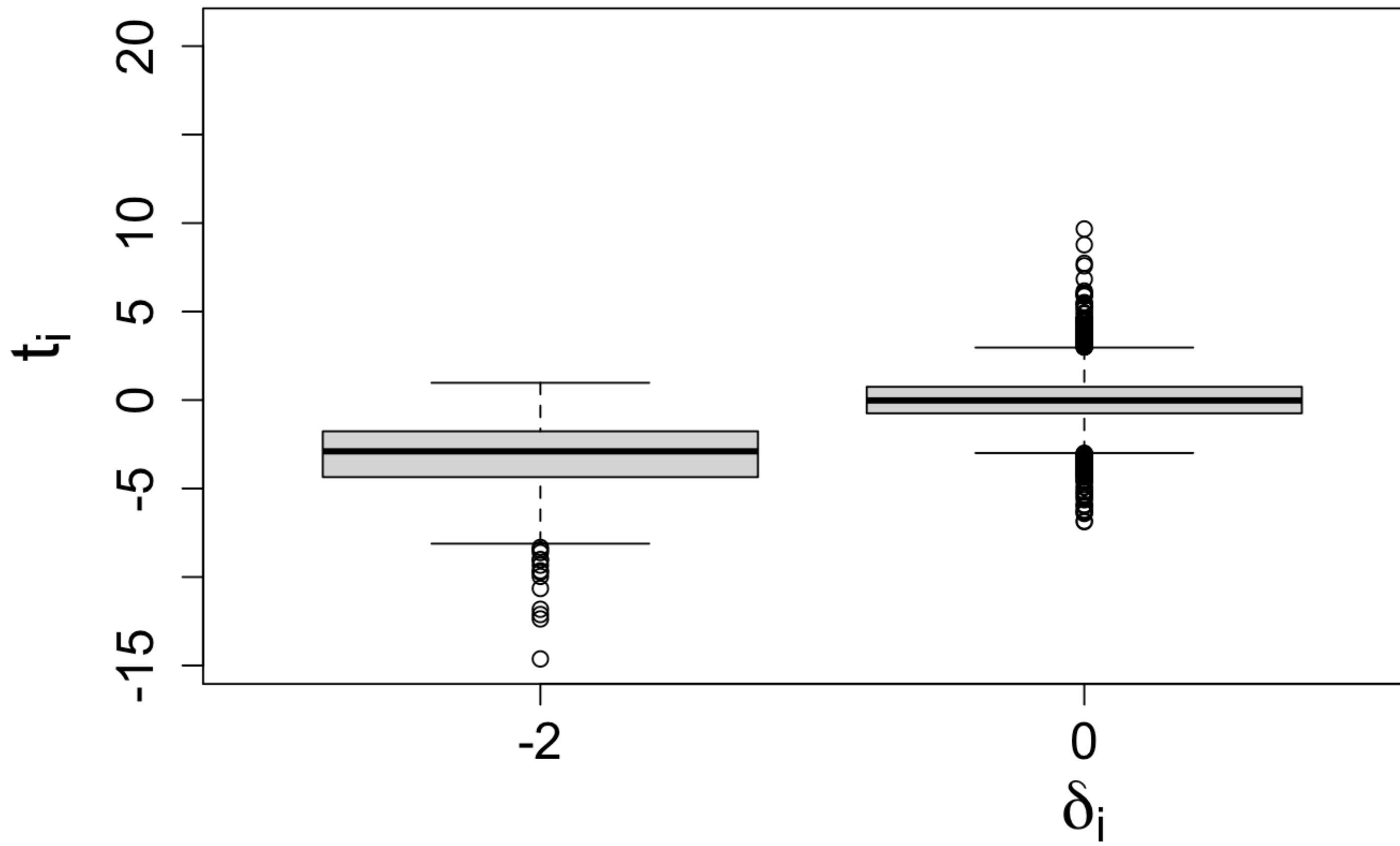
$$X_{ij} \sim N(\mu_{ij}, \sigma_i)$$
$$\mu_{ij} = \mu_{i0}, \quad j \in A$$
$$\mu_{ij} = \mu_{i0} + \delta_i, \quad j \in B$$

- N = 5000, M = 6
- $\delta_i = 0$ for 90%
- $\delta_i = \pm 2$ for 10%
- $\sigma_i \sim \Gamma(10, 10)$

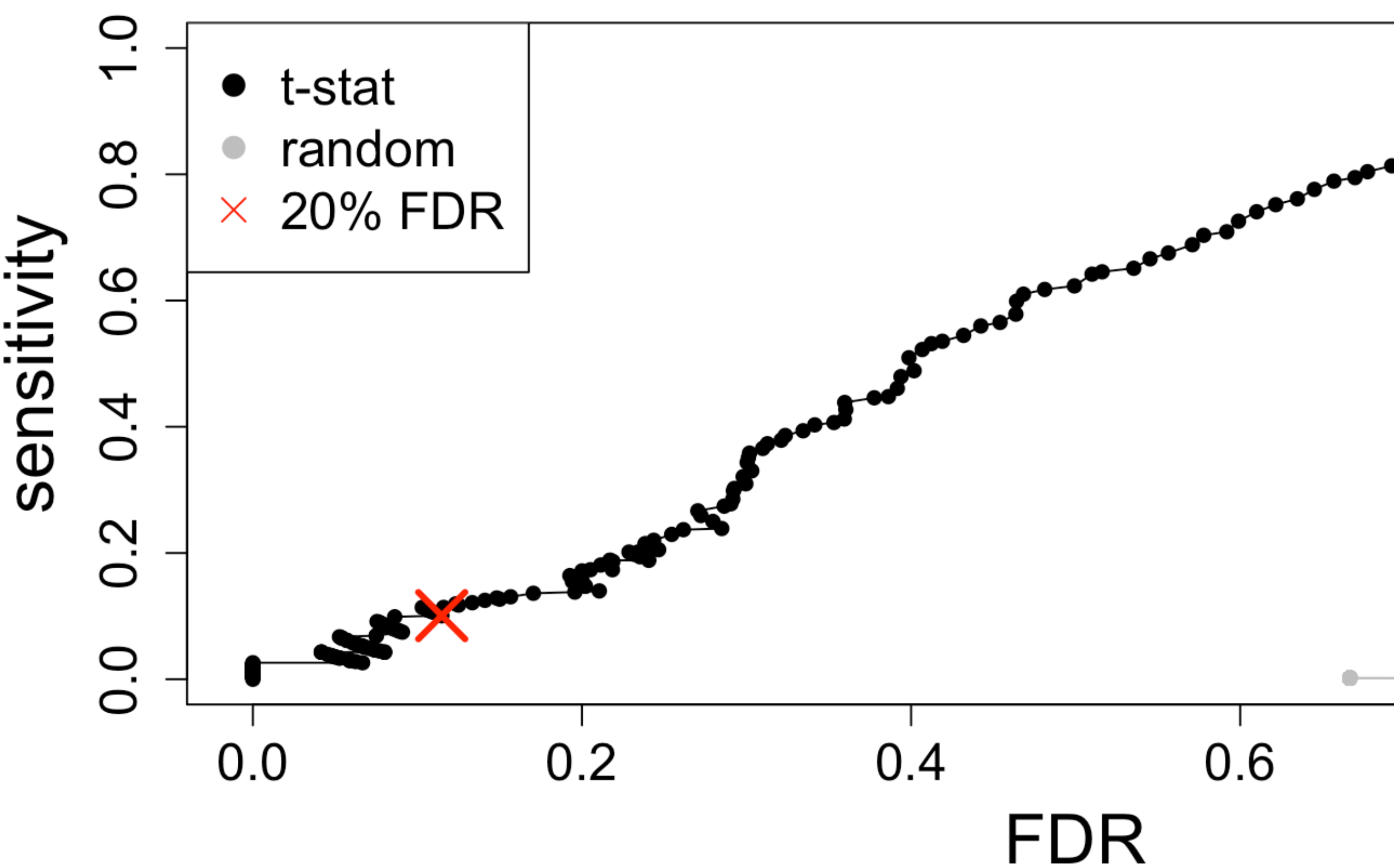
Distribution of σ_i



Try simple row t-tests

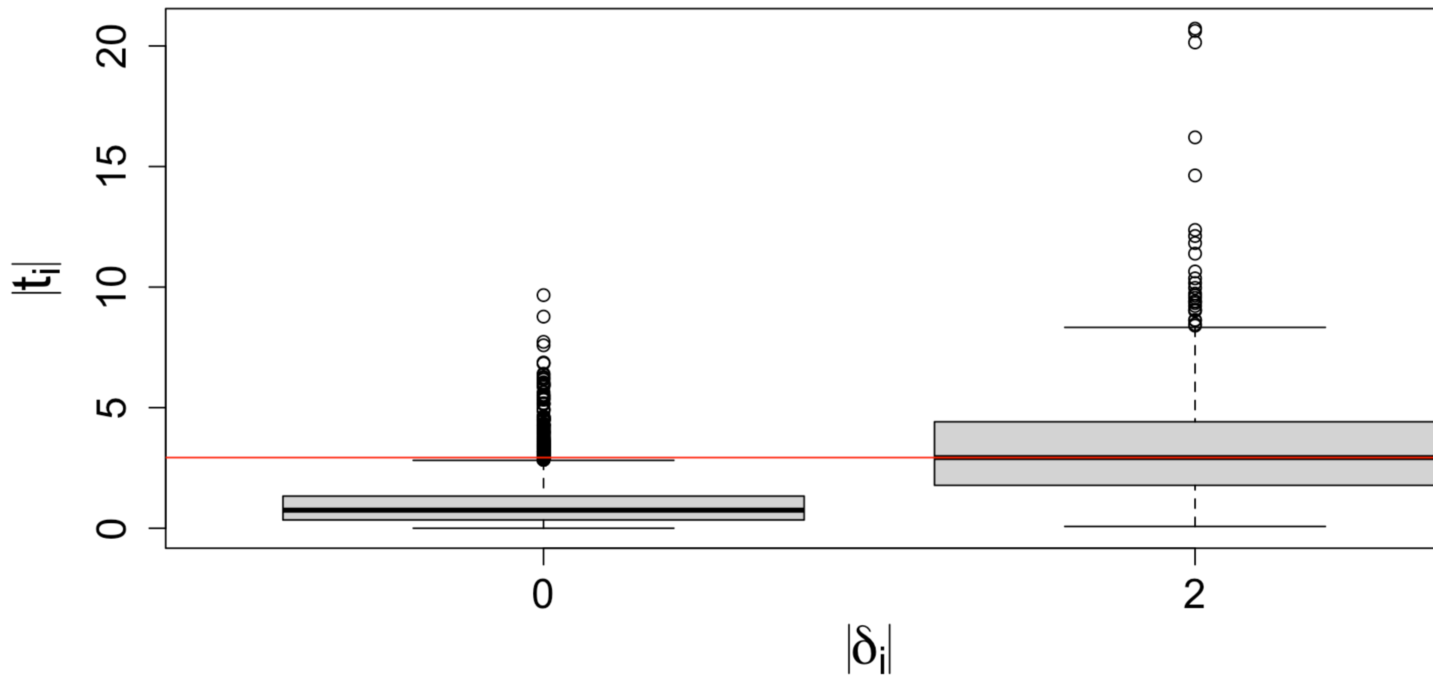


Just looking at ranks



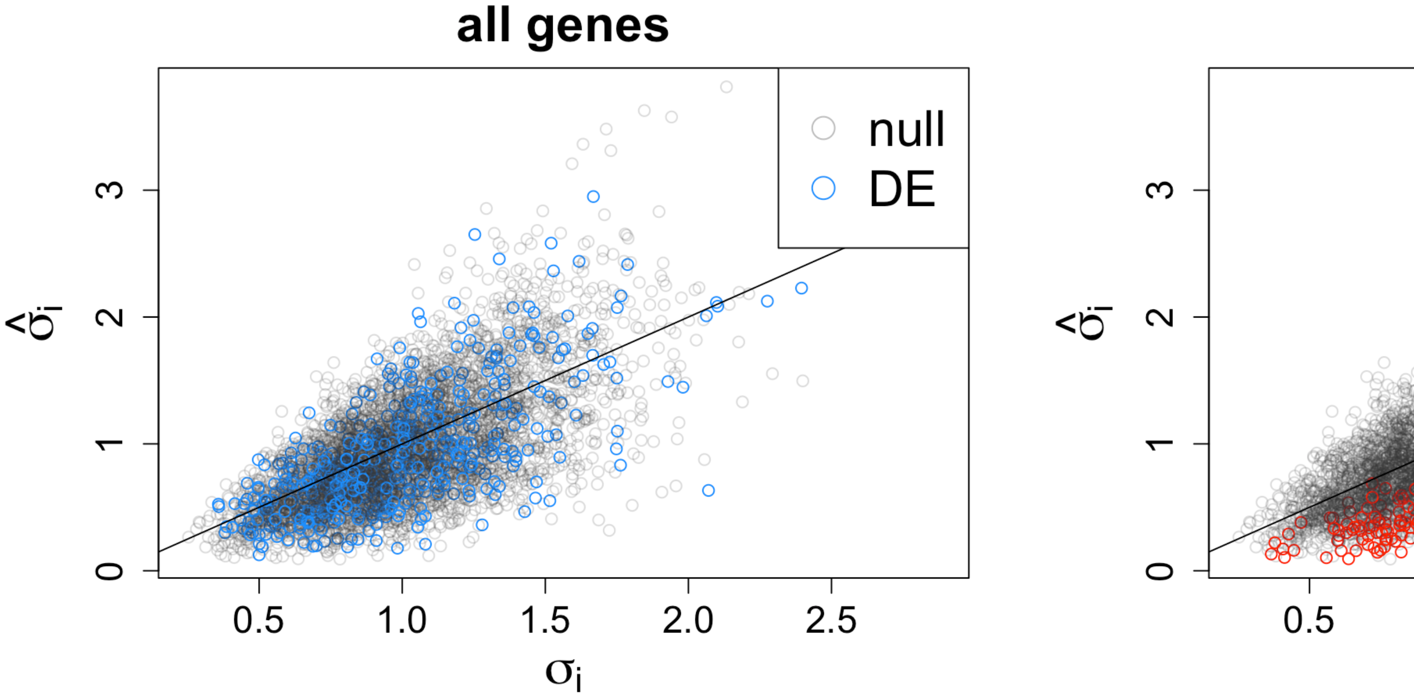
Characterize the false positives

$$\mathrm{med}(t) \equiv \mathrm{median}(|t_i|)\ \mathrm{for}\ i :$$



Estimates of σ_i

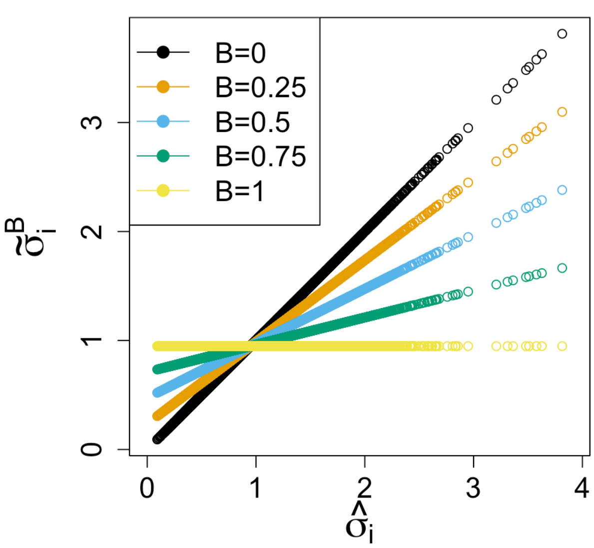
$$\mathrm{med}(t) \equiv \mathrm{median}(|t_i|)\,\mathrm{for}\, i :$$



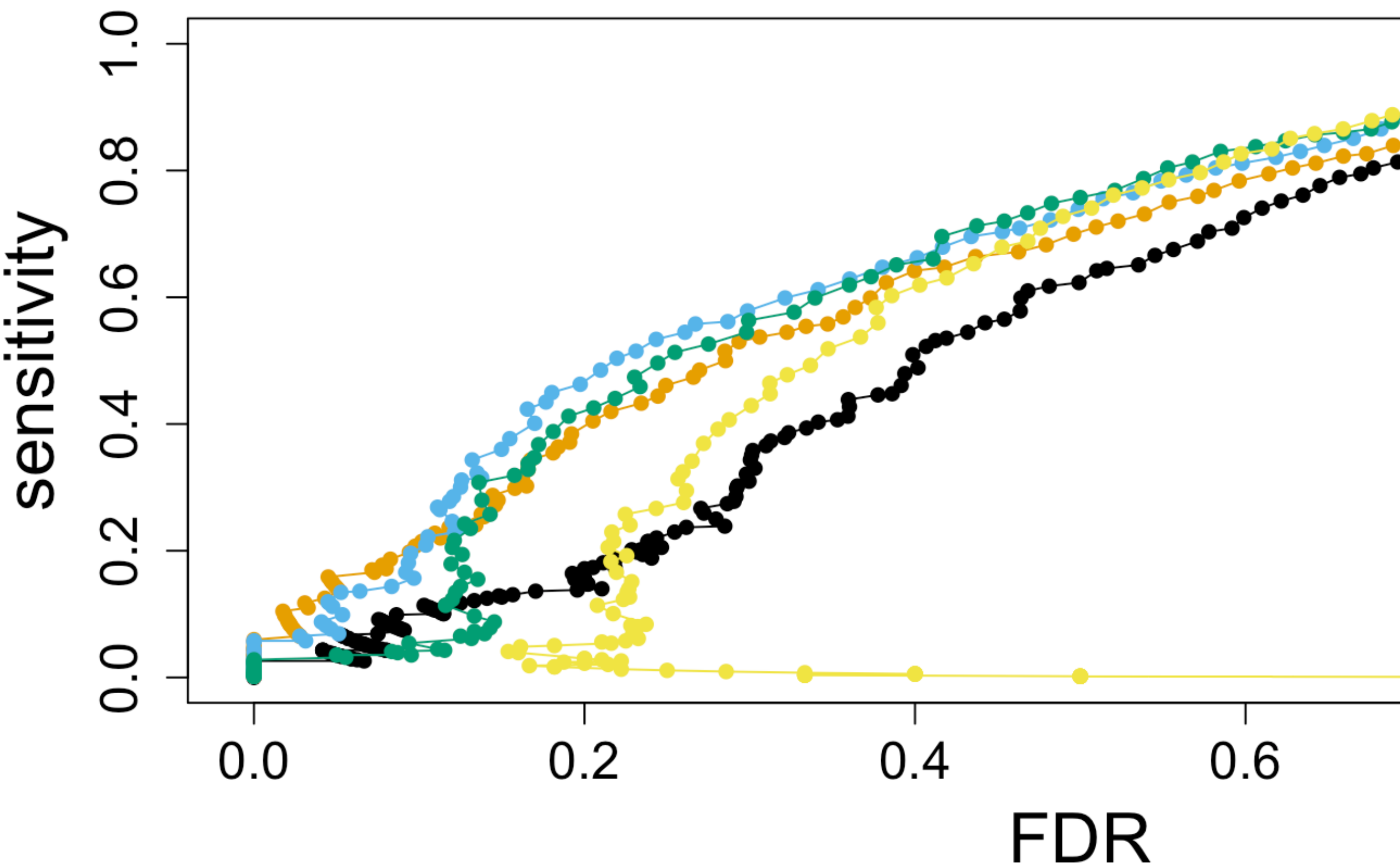
New estimator for σ_i

$$\bar{\sigma} = \frac{1}{N}\sum_{i=1}^{N}\hat{\sigma}_i$$

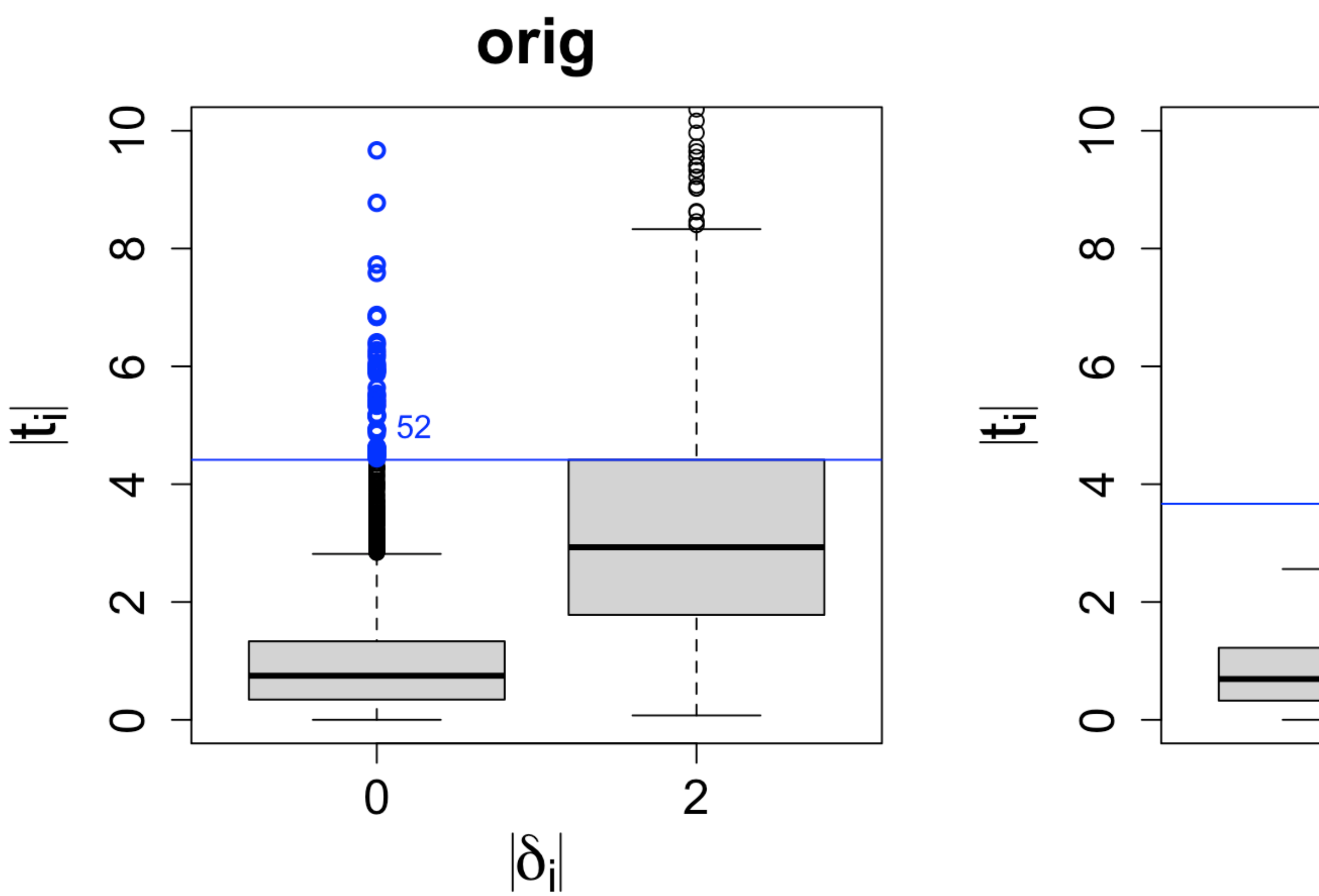
$$\tilde{\sigma}_i^B = B\bar{\sigma} + (1-B)\hat{\sigma}_i$$



New estimator performance by r



New estimators by |t|

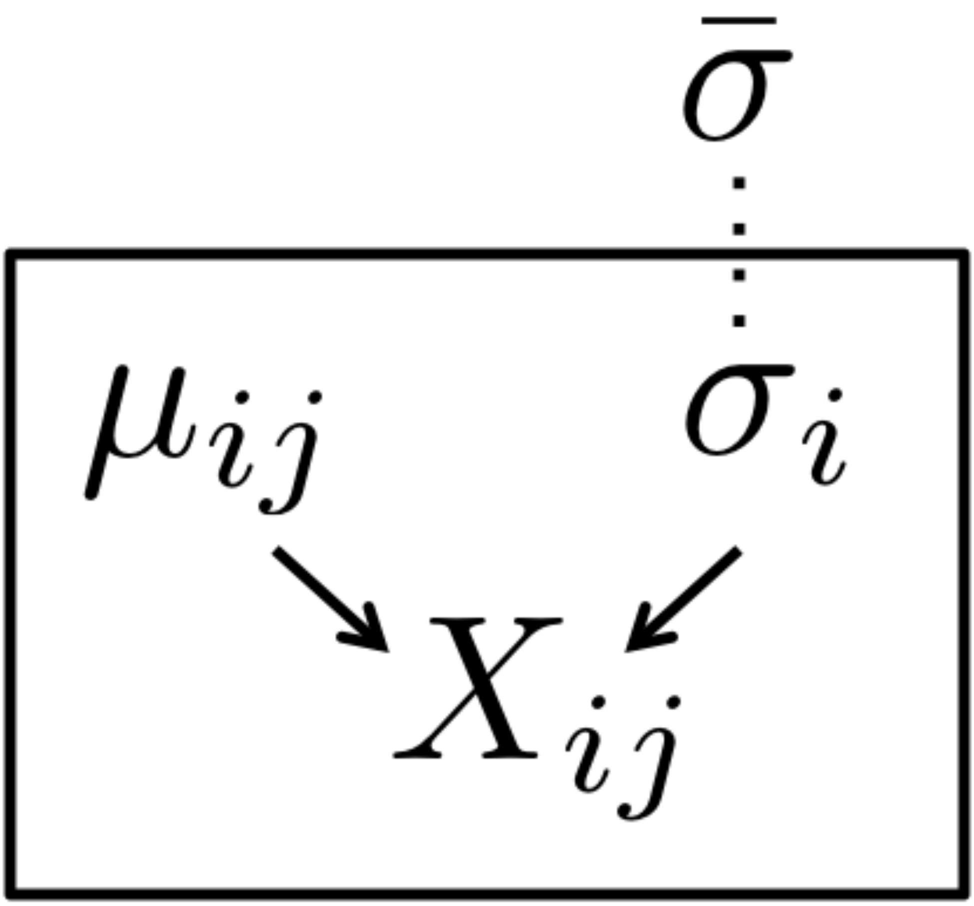


Summary

- Top false positives were coming from ger
- Replace $\hat{\sigma}_i$ with an estimate which is clos
- Depending on “close”, new estimator dor thresholds

How is this hierarchical?

Not your standard diagram, need to formali



limma

- Smyth, G. K. (2004) Linear models and em
methods for assessing differential expres
experiments
- Developed the hierarchical model introdu
and Speed (2002) for single sample into r
experiment represented as linear model

$$\frac{1}{\sigma_i^2} \sim \frac{1}{d_0 \sigma_0^2} \chi_{d0}^2$$

Why inverse χ^2?

- Conjugacy provides closed form solution
- Posterior mean for $1/\sigma_i^2$ given $\hat{\sigma}_i^2$ is $1/\tilde{\sigma}_i^2$

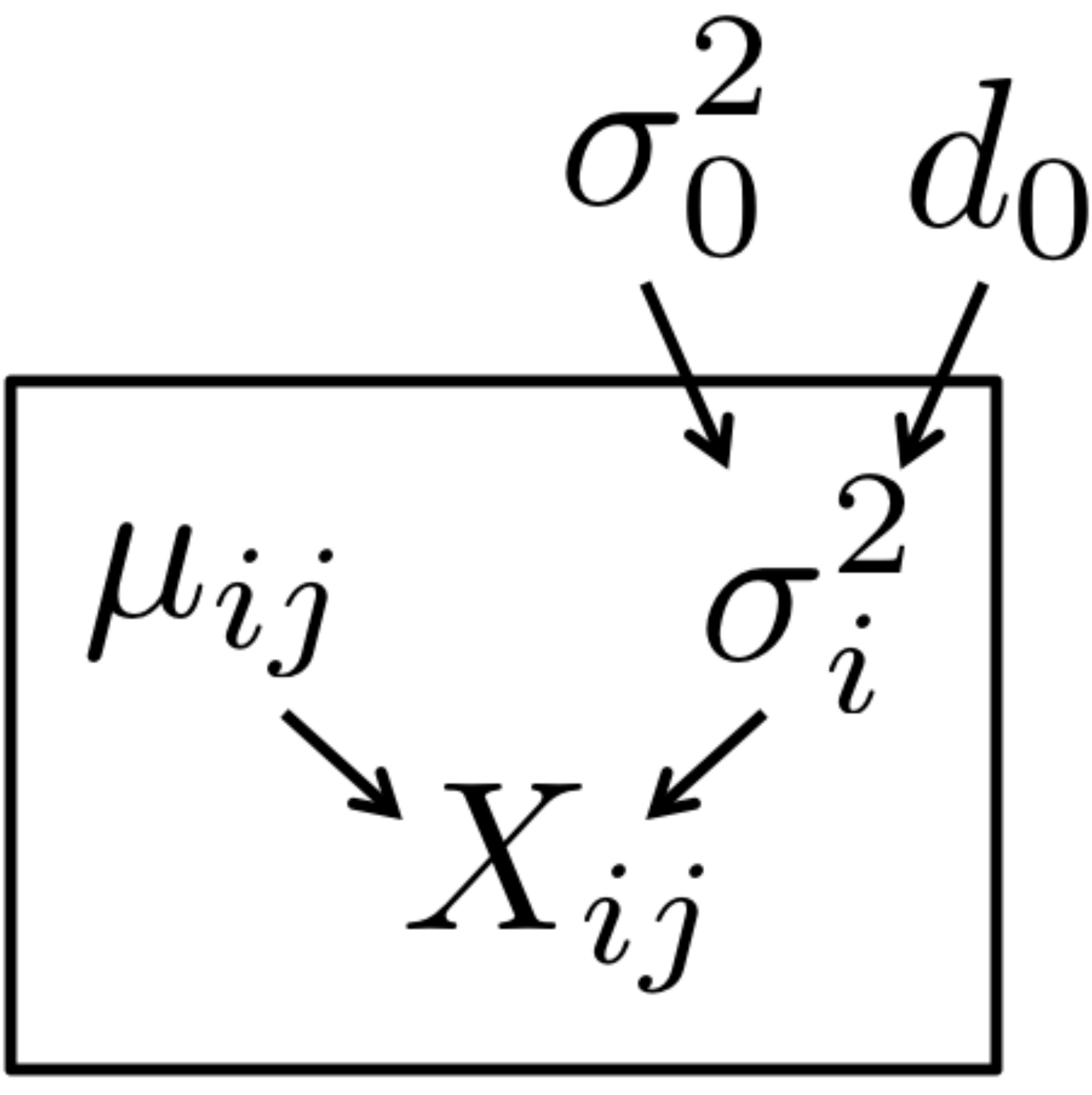
$$\tilde{\sigma}_i^2 = \frac{d_0\hat{\sigma}_0^2 + d_i\hat{\sigma}_i^2}{d_0 + d_i}$$

And d_i as the standard residual degrees of f

Note that d_0 controls B

$$
\begin{aligned}
\tilde{\sigma}_i^2 &= \frac{d_0 \hat{\sigma}_0^2 + d_i \hat{\sigma}_i^2}{d_0 + d_i} \\
&= \left(\frac{d_0}{d_0 + d_i} \right) \hat{\sigma}_0^2 + \left(\frac{d_i}{d_0 +} \right. \\
&= B \hat{\sigma}_0^2 + (1 - B) \hat{\sigma}_i^2
\end{aligned}
$$

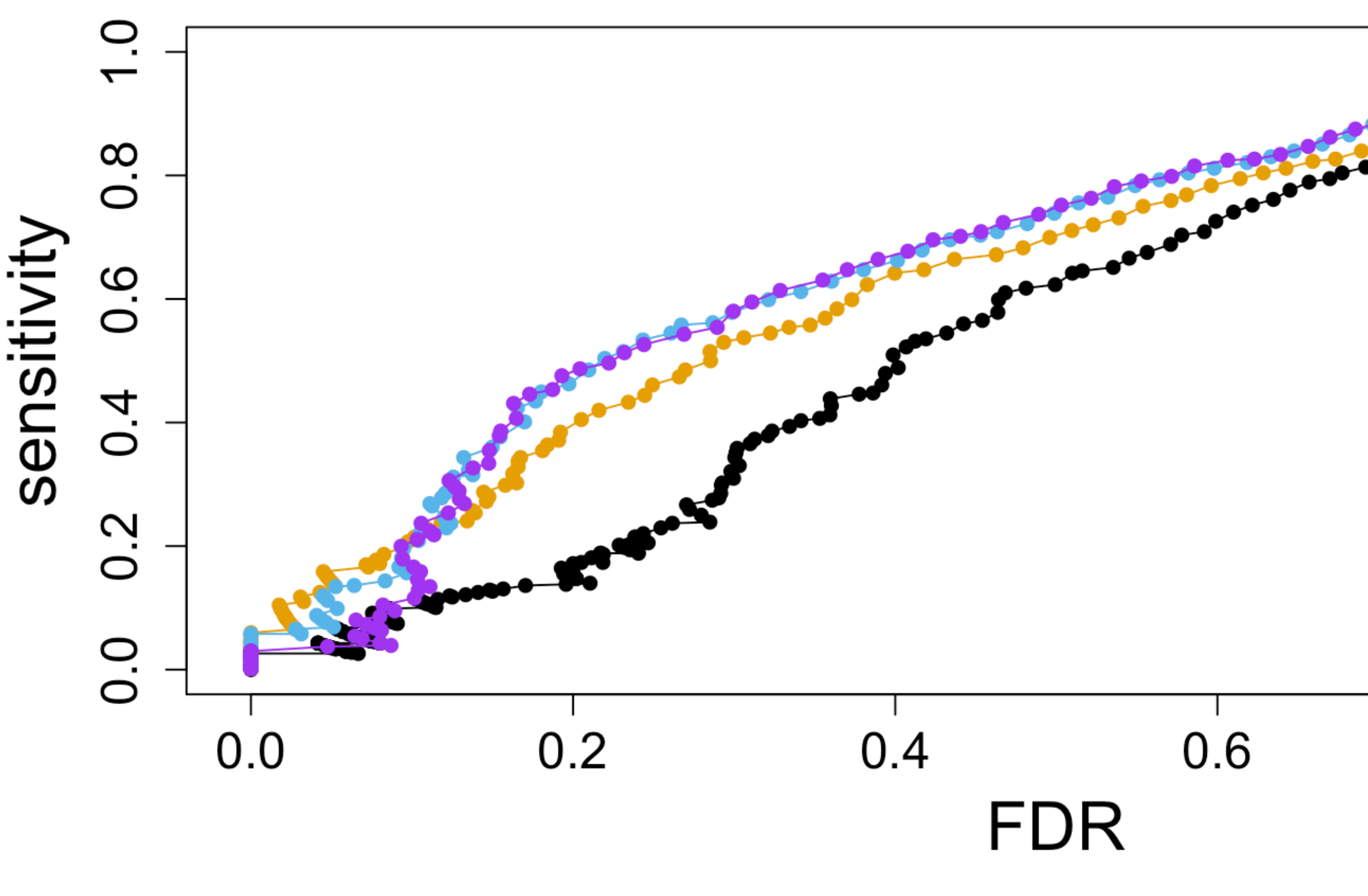
Proper hierarchical model



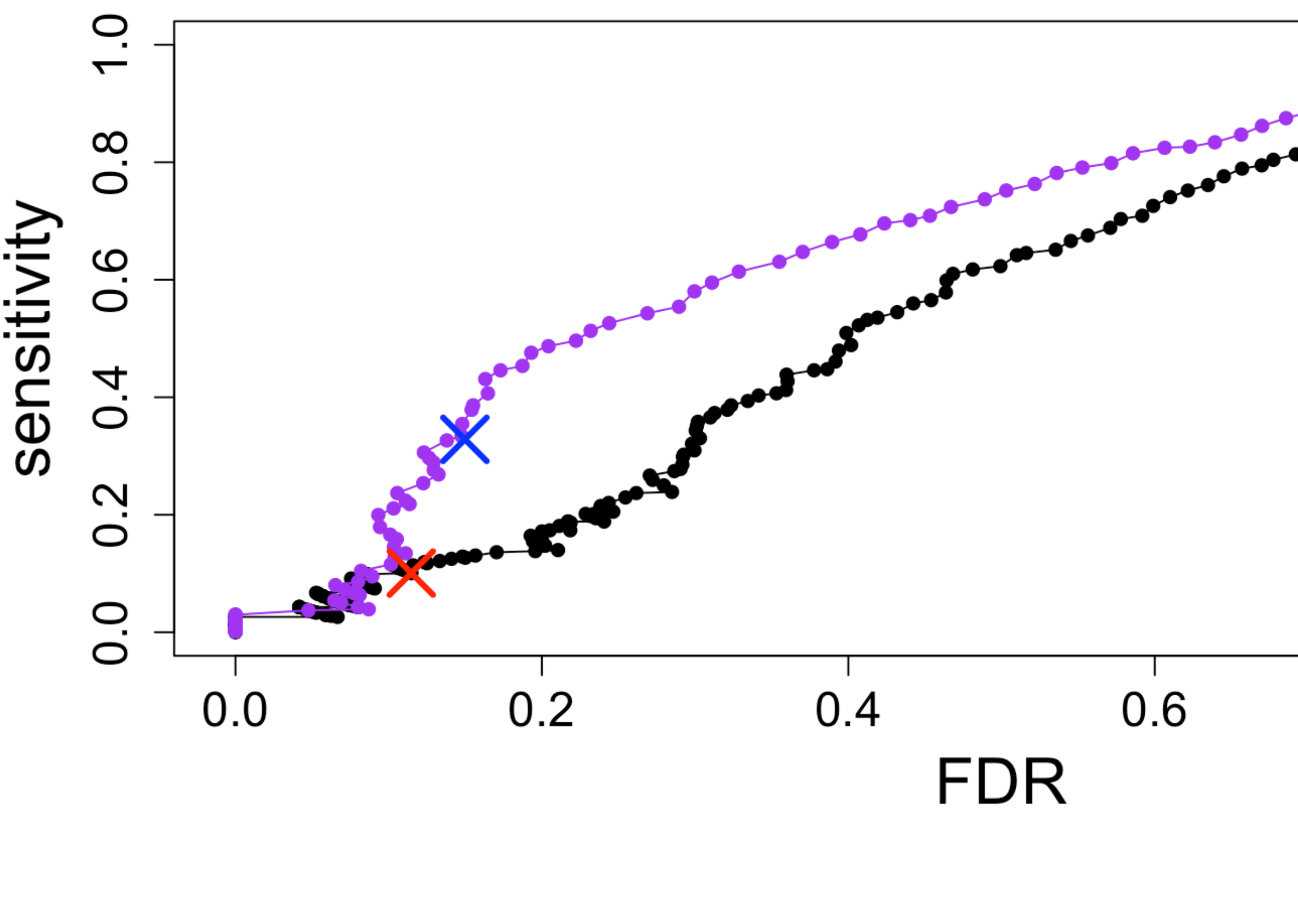
Estimation of hyperparameters

- Need to estimate $d_0, \hat{\sigma}_0^2$, which control st
  of *shrinkage* or *moderation*
- $d_0, \hat{\sigma}_0^2$ estimated via first two moments o
- (Also need to estimate v_0, another param
  variance of coefficients)

limma vs. naive estimators by ra



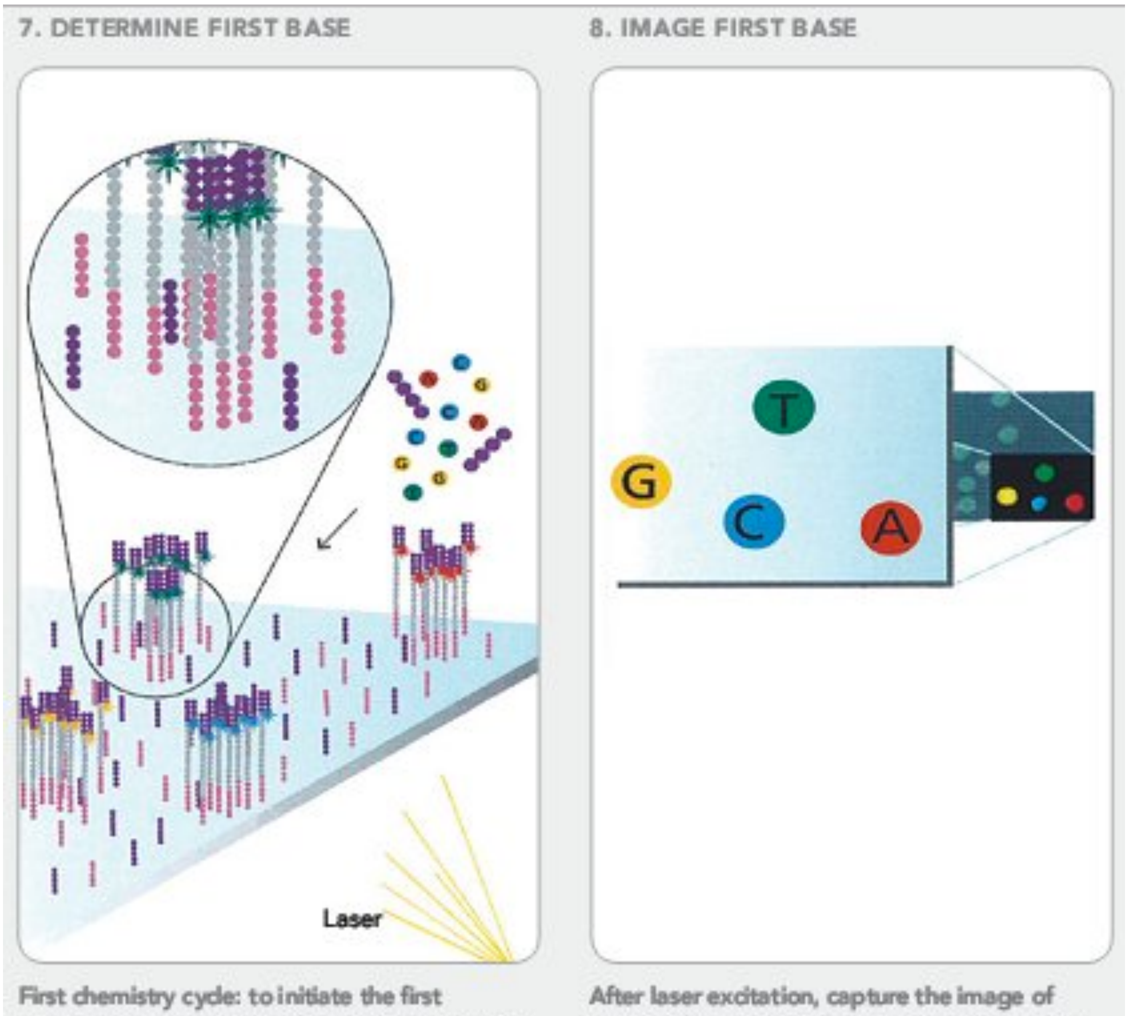
Operating characteristics (FP & F



Summary

- limma provides a hierarchical model for r
  variance estimates in the context of linea
- Avoids false positives from under-estimat
- Also addresses the gain in degrees of free
  moderation

RNA-seq: counting molecules



First chemistry cycle: to initiate the first sequencing cycle, add all four labeled reversible terminators, primers and DNA polymerase enzyme to the flow cell.

After laser excitation, capture the image of emitted fluorescence from each cluster on the flow cell. Record the identity of the first base for each cluster.

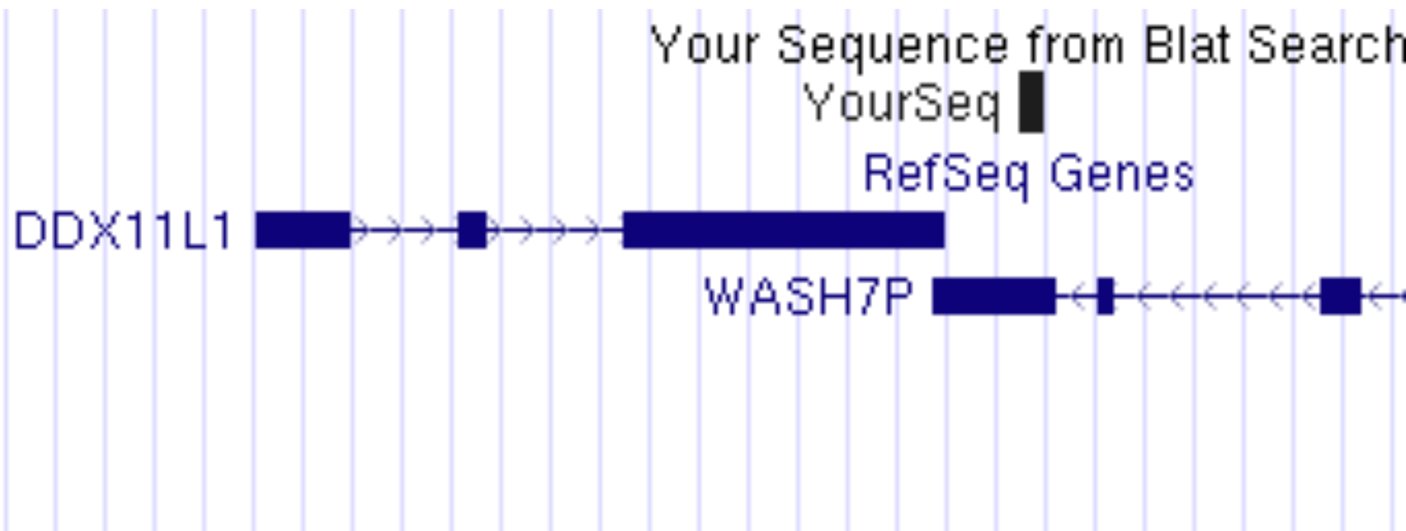
RNA-seq: counting molecules

```
@SRR1265495.1 1/1
CTTTGCCCGCGTGTCAGACTCCATCCCTCCTCTGCCGCCACCGCAGCAGCCACA
+
CCCFFFFFHGFHHHIJJIJIJJIJJJJJJGIIJJJIJJJJJJJEFHDEFFFE
B@BDDD@;9<BBBDBCB@A2<?BDDDBDBD@CDDC
@SRR1265495.2 2/1
CCTGGCTGTGTCCATGTCAGAGCAATGGCCCAAGTCTGGGTCTGGGGGGGAAGG
+
@C@FFFDFHHHHGGIIAGHI9GIIIIIGIIIIIGI@@FHGDDDH@GGIBB05?B
BBCCBBBBCBCDC3>AC<BB<?CBBB?<@CCCA:@
@SRR1265495.3 3/1
CTGTGTCCATGTCAGAGCAATGGCCCAAGTCTGGGTCTGGGGGGGAAGGTGTCA
+
@C@DFFFFGGHDHIIEGDHCGGHIJGEHIIJIIJIEEHGIGEGDDB@@@CDDDD
```

for ~30 million reads (often pairs of reads)

RNA-seq: counting molecules

Align to genome or transcriptome



RNA-seq: counting molecules

- Now for each gene and each sample we c
  or *estimated count* of fragments
- Why estimated? Because some fragments
  uniquely associated with genes or isoforn
- Fast algorithms for probabilistically assig
  - kallisto (2016)
  - salmon (2017)
- Assume we have integer counts K_{ij} of ur
  from rounding estimated counts

Counts

1. Either model data with count distributio
   with GLM

- DESeq2 (2014)
- edgeR (2010)
- many more

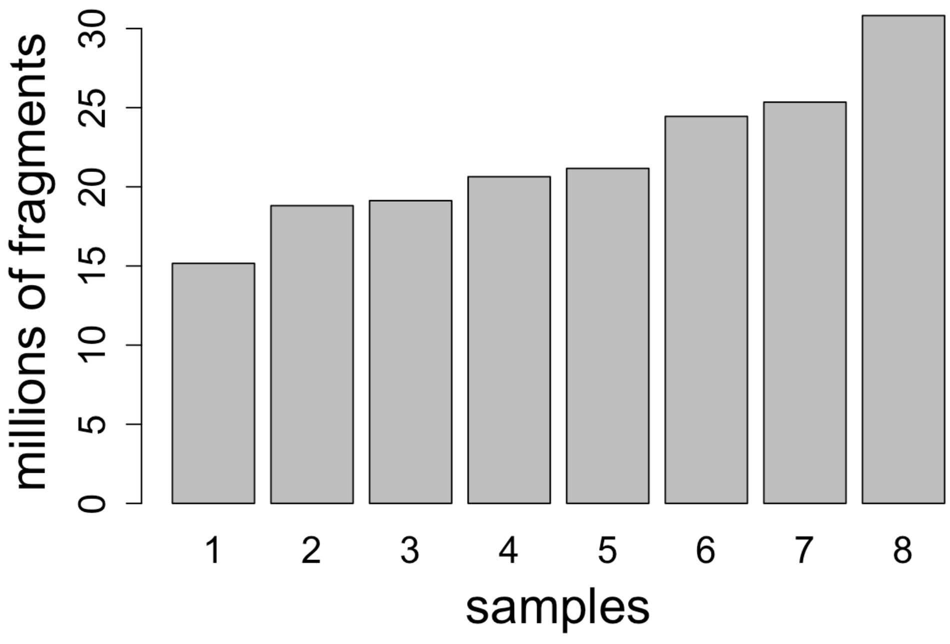
2. Learn weights associated with log norma
   use limma

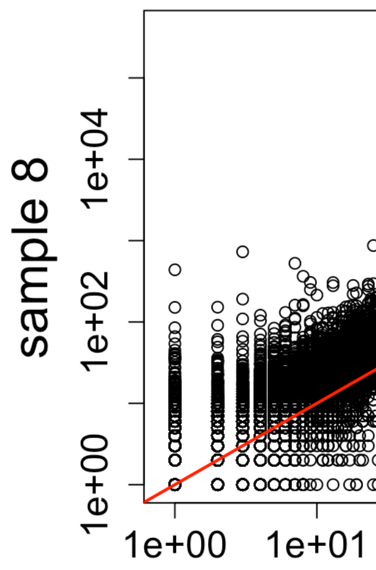
- limma-voom (2014)

Important for statistical analysis

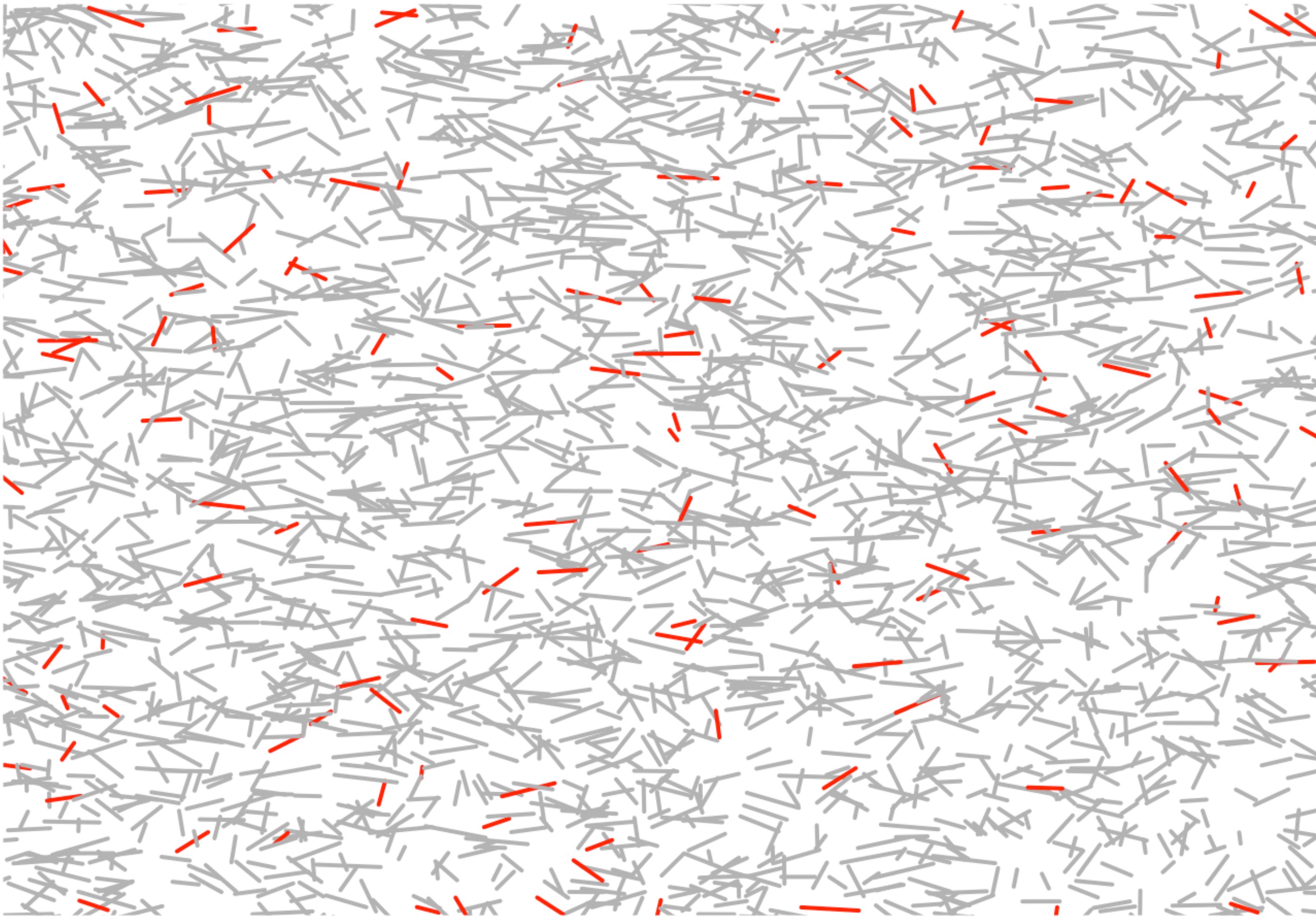
- Total number of fragments is technical ar
- Heteroskedasticity of counts
- Each gene has different variability
- Technical covariates can not be excluded
- With $n > 20$, estimate and include techn design

Total number of fragments



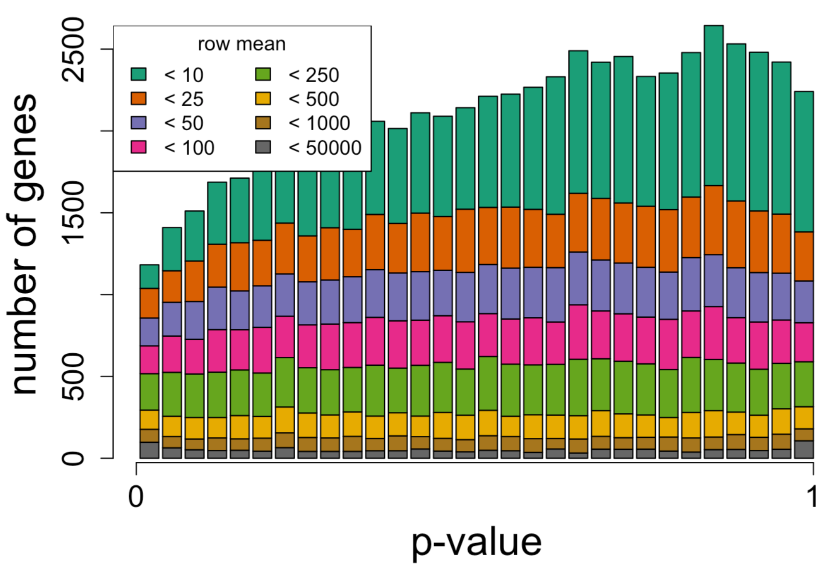


Sampling fragments



Poisson across technical replicat

- From Bullard 2010 take 7 technical replic
- Calculate expected value $\hat{\lambda}_{ij}$ using DESeq
- $P(K_{ij} < \hat{\lambda}_{ij})$ assuming $K_{ij} \sim \mathrm{Pois}(\hat{\lambda}_{ij})$

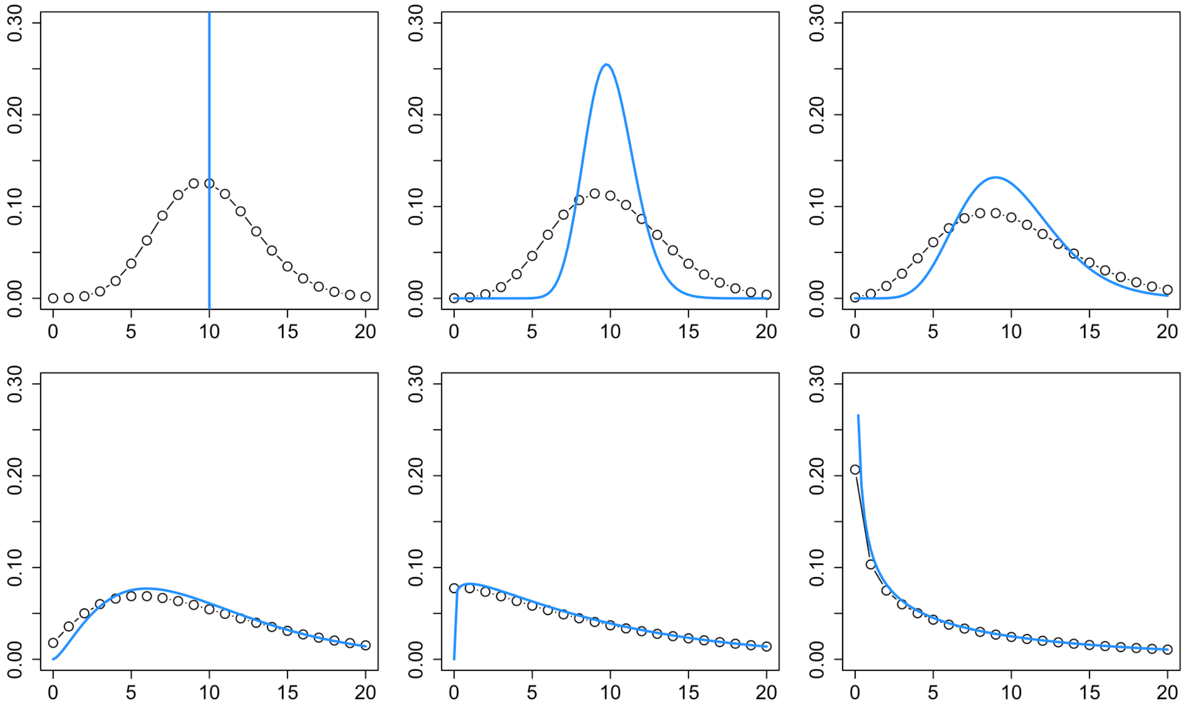


However, expression not equal a
biological replicates

Negative Binomial / Gamma Pois

$$K_{ij} \sim \mathrm{NB}(\mu_{ij}, \alpha_i$$

$$\mathrm{Var}(K_{ij}) = \mu_{ij} + \alpha_i \mu_{ij}^2$$



NB model for RNA-seq

- s_j deals with sequencing depth

$$
\begin{aligned}
K_{ij} &\sim \mathrm{NB}(\mu_{ij}, \alpha_i) \\
\mu_{ij} &= s_j q_{ij} \\
\log_2(q_{ij}) &= \beta_{i0}, \quad j \in A \\
\log_2(q_{ij}) &= \beta_{i0} + \delta_i, \quad j \in
\end{aligned}
$$

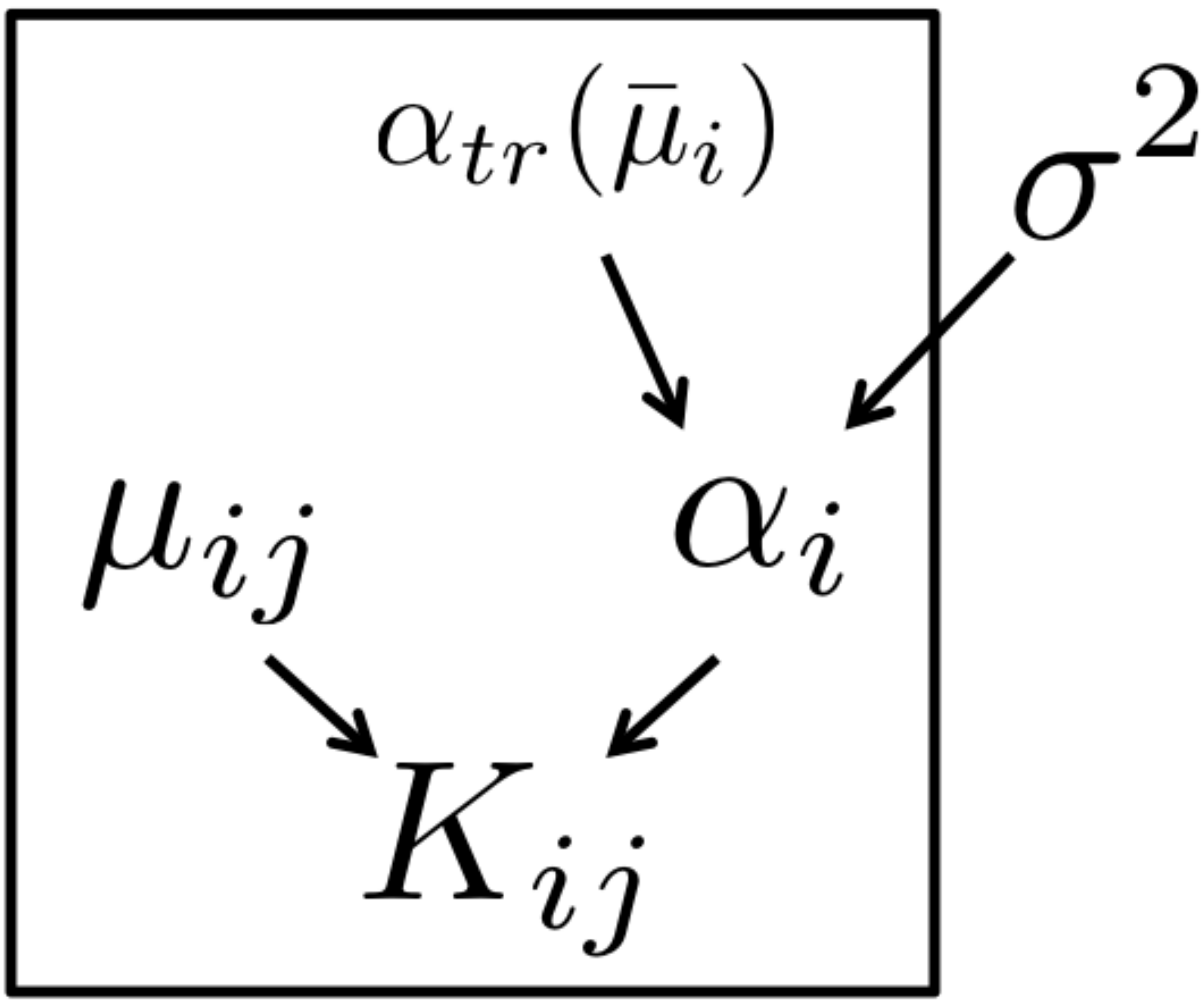
$\delta_i \neq 0$ implies DE (differential expression)

Moderation of dispersion

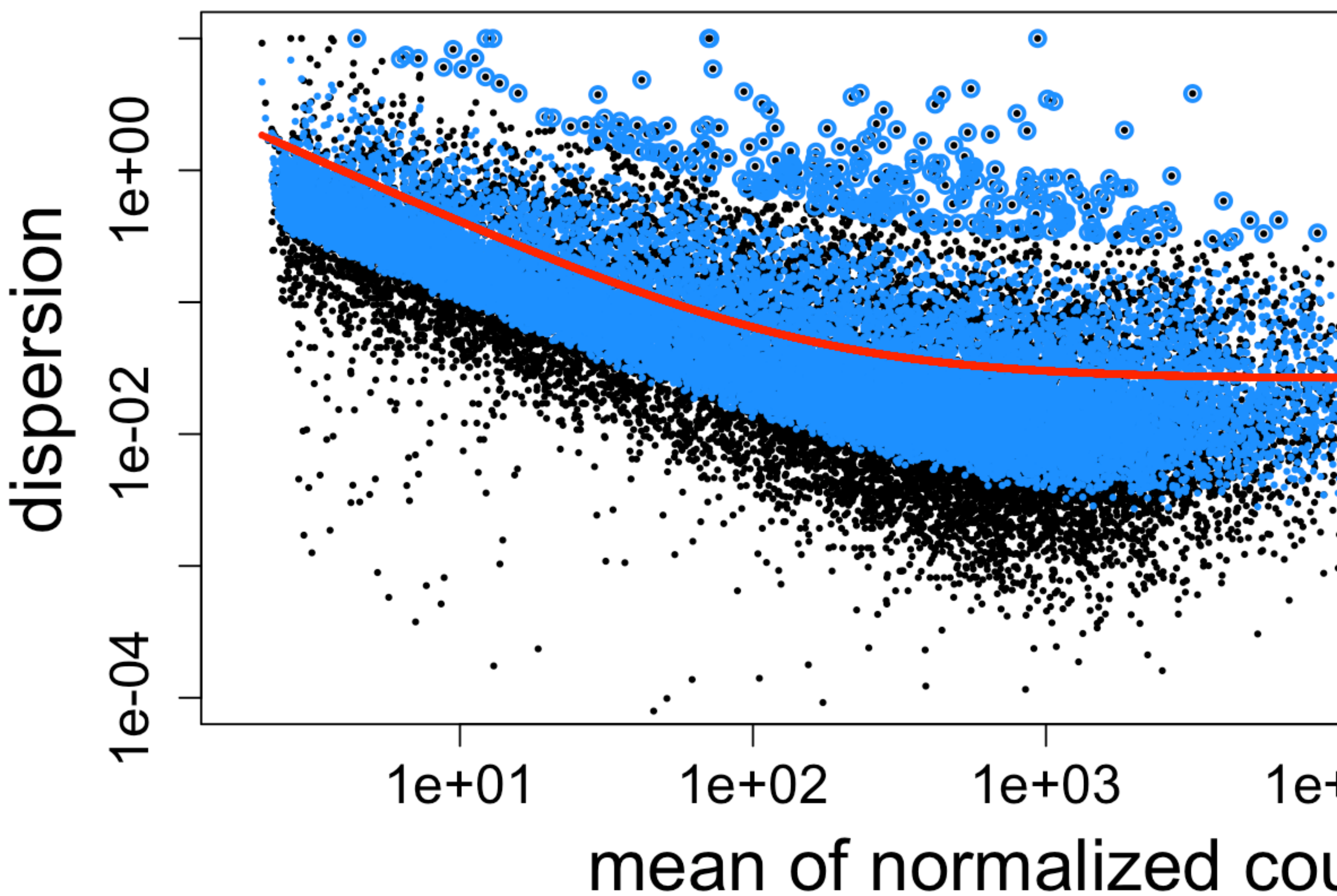
- In DESeq2, a prior on $\log(\alpha_i)$
- Calculate the mean of normalized counts
- A trend line of dispersion over mean $\alpha_{tr}($
- Width of the prior σ^2 estimated via assur
sampling variance of $\log(\hat{\alpha}_i)$

$$\log(\alpha_i) \sim N(\log(\alpha_{tr}(\bar{\mu}_i)),$$

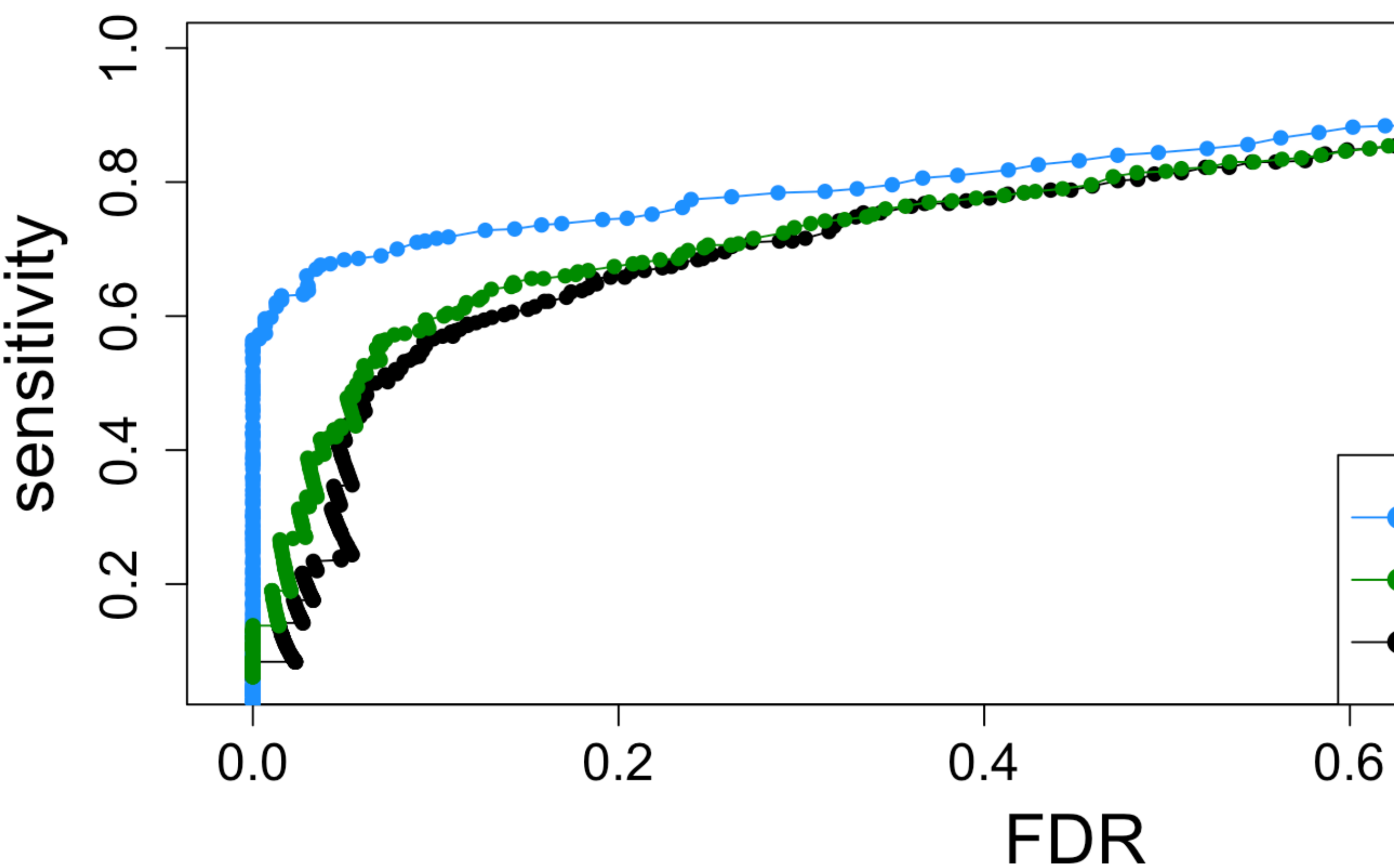
Moderation of dispersion



Moderation of dispersion



Evaluate via simulation



Summary

- Model for counts similar to the hierarchic
  constructed for the dispersion parameter
- Final dispersion estimate plug-in value in
- edgeR quasi-likelihood takes into accoun
  estimation uncertainty
- limma-voom uses weights on log normal
- apeglm uses Cauchy prior (heavy tails) or

Questions?